

Advanced signal processing - Allpass and invertibility problems

1 Allpass filters and reverberators

The goal of this problem is to try out how two different reverberation algorithms sounds in practice.

1. Read example 5.7 and 5.9 in Manolakis and Ingle. In particular, pay attention to the two impulse responses (Fig. 5.26 and 5.32) and compare them.
2. Find a way of implementing the reverberators in MATLAB.
3. Assume we will use the reverberators on a 48 kHz system. What are suitable values of a and D ?
4. On Brightspace you can find a short piece of music sampled at 48 kHz. Use `wavread` to load this into MATLAB and play it with `audioplayer`.
5. Run the music through the two reverberators and listen to the output. Is it possible to tell the difference?

2 Cascading all-pass filter

The transfer function of a 1st order all-pass filter is given by

$$H_{AP}(z) = \frac{z^{-1} - p^*}{1 - pz^{-1}}$$

1. Show that the cascade of two 1st order all-pass filters having complex conjugated poles, $p_1 = p_2^*$ gives a second order all-pass filter with real coefficients.

3 Quantization effects in all-pass filters

An All-Pass filter has the following system function

$$H(z) = \frac{0.57 + 0.23z^{-1} + z^{-2}}{1 + 0.23z^{-1} + 0.57z^{-2}}$$

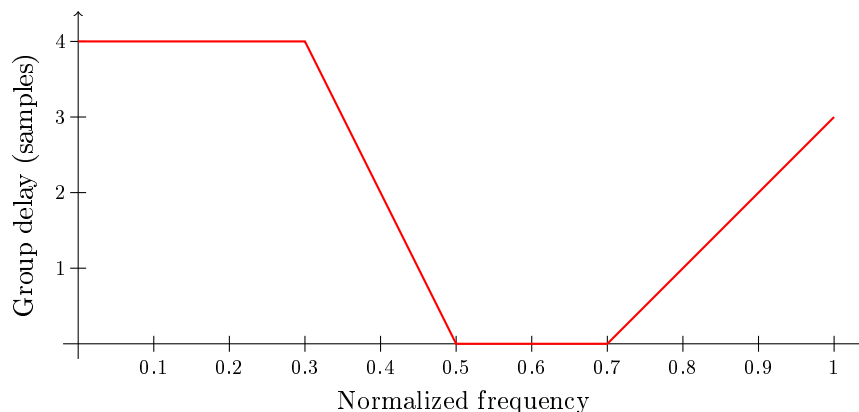
1. First, without performing calculations, try to predict what the impulse response of the filter will look like. Second, plot the impulse response and compare with your prediction.

Next, assume that the filter is to be implemented on a fixed point 16 bits signal processor, e.g. an Analog Devices Blackfin. This implies that the coefficients must be quantized to (1.15) format.

2. Is the filter still an all-pass filter when the coefficients have been quantized ?
3. If you recall how the calculation works, quantize the filter coefficients to 16 bits in (1.15) format.

4 Designing all-pass filters

Assume it is desired to create an all-pass filter with group delay versus frequency as shown in plot below. Due to the sharp edges in the group delay such a filter can hardly be designed but an approximation is feasible.



1. Use `iirgrpdelay` in MATLAB to design a 4th order all-pass filter. You can set `edges=[0 1]`.
2. What happens as you increase the order of the filter?
3. What order of the filter is necessary to bring the maximum group delay error below 0.5 samples?
4. Plot the impulse response of the filter. Does the length of the impulse response match your expectations.

5 Filter decomposition

Let the system function for a FIR filter be given by

$$H(z) = 1 - 3z^{-1} + \frac{5}{2}z^{-2} - z^{-3}$$

1. Decompose the system function into a product of a minimum-phase filter and an all-pass filter, $H(z) = H_{min}(z)H_{ap}(z)$.
2. Demonstrate that $H(z)$ and $H_{min}(z)$ have the same amplitude response.
3. Repeat for the following filter

$$H(z) = \frac{1 + \frac{3}{2}z^{-1} - 9z^{-2} + 4z^{-3}}{1 - \frac{1}{2}z^{-1}}$$

4. What are the real world applications of this kind of filter decomposition ?

6 Filter decomposition

Consider a FIR filter with the following difference equation

$$y(n) = x(n) + 2x(n-1) + 2x(n-2)$$

1. Show that the FIR filter is not minimum-phase.
2. Find the difference equation for the corresponding minimum-phase FIR filter.

7 Minimum, mixed and maximum phase filters

Consider a FIR filter with two real zeros, $z_1 = 0.5$ and $z_2 = -\frac{1}{3}$.

1. Calculate the transfer function $H_1(z)$ for the filter and argue that it is minimum phase.
2. Calculate the possible mixed phase and maximum phase filters for FIR filters with the same amplitude response.
3. Plot the difference in group delay for the minimum phase and maximum phase filters.

8 Group delay in FIR filters

Consider a FIR filter with a single zero located at $z_0 = re^{j\theta}$. This means that

$$\begin{aligned} H(\omega) &= 1 - re^{j\theta}e^{-j\omega} \\ &= 1 - r\cos(\omega - \theta) + jr\sin(\omega - \theta) \end{aligned}$$

1. Show that the phase response is

$$\Psi(\omega) = \tan^{-1} \left(\frac{r \sin(\omega - \theta)}{1 - r \cos(\omega - \theta)} \right)$$

2. Show that the group delay becomes

$$\tau(\omega) = \frac{r^2 - r \cos(\omega - \theta)}{1 + r^2 - 2r \cos(\omega - \theta)}$$

3. What physical unit is used to measure group delay ?

Assume that the signal in a 48 kHz audio system is filtered with a FIR filter with the impulse response $h(n) = \{1, \frac{1}{2}\}$.

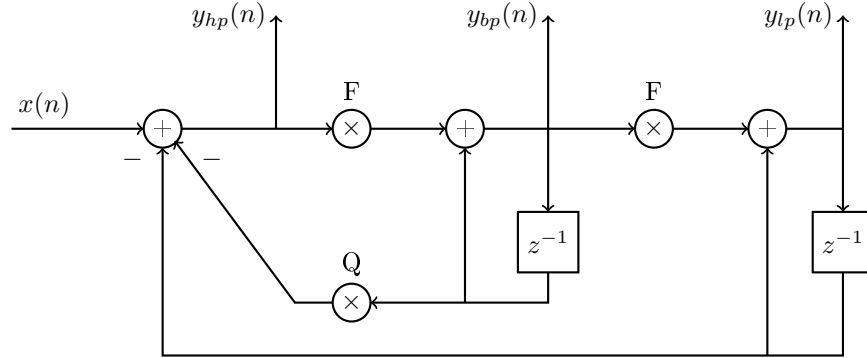
4. Use the results from the first part of the problem to plot the group delay, in units of μs , in the filter as a function of the frequency measured in Hz. MATLAB's `grpdelay` command can be used to verify your results.

Mathematical hints for the problem:

$$\cos^2(x) + \sin^2(x) = 1, \quad \frac{d}{dx} \tan^{-1}(u) = \frac{1}{1+u^2} \frac{du}{dx} \quad \frac{d}{dx} \left(\frac{f(x)}{g(x)} \right) = \frac{\frac{df(x)}{dx} g(x) - f(x) \frac{dg(x)}{dx}}{g^2(x)}$$

9 The digital state variable filter

One interesting filter structure is the digital state variable filter which has its roots in the analog state variable filter. It can be considered as a standard second order filter, but with the valuable feature that the center frequency and Q -factor are independently tuneable. The structure of the filter is shown here



Where $F = 2 \sin(\pi F_c / F_s)$ with F_c the filter corner frequency, F_s is the sampling frequency and Q is the Q -factor of the filter. From the structure, updating equations for the three state variables, $y_{hp}(n)$, $y_{bp}(n)$ and $y_{lp}(n)$ can easily be written down. Index hp, bp and lp denotes highpass, bandpass, and lowpass respectively. For $y_{hp}(n)$ the equation becomes

$$y_{hp}(n) = x(n) - y_{lp}(n-1) - Qy_{bp}(n-1).$$

1. Write down the updating equations for the two other state variables.
2. Calculate the transfer function $H_{LP}(z)$ from the input to the lowpass output.
3. Implement the state variable filter as a function in MATLAB. Make sure the three equations are updated in the correct order.
4. Test the implementation by sending white noise sampled at 48000 Hz through the filter. Use $F_c=1000$ Hz and $Q=2$. Plot the spectra of the three outputs on the same graph and compare with the expected spectra. Repeat for $Q=10$.

The filter is described in P. Dutilleul "Simple to operate digital time varying filters" AES 86th Convention, Hamburg (1989) In this paper Dutilleul shows that the digital state variable filter fails if F_c becomes too large. A rule of thumb is to keep $F_c < F_s/10$.

5. Demonstrate this is true by using $F_c=20000$ Hz in the above implementation.
6. A band reject output is also easily obtained with the digital state variable filter. Extend the above drawing with a $y_{br}(n)$ output.